

Intergalactic Riksbanken Chip Authenticator - Screenshots

Group 8

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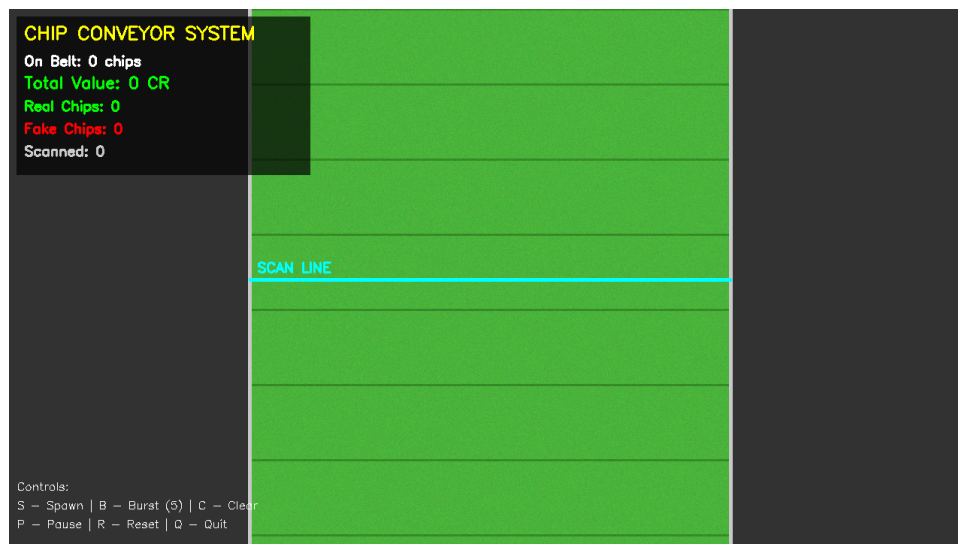
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Project Screenshots

Intergalactic Riksbanken Chip Authenticator

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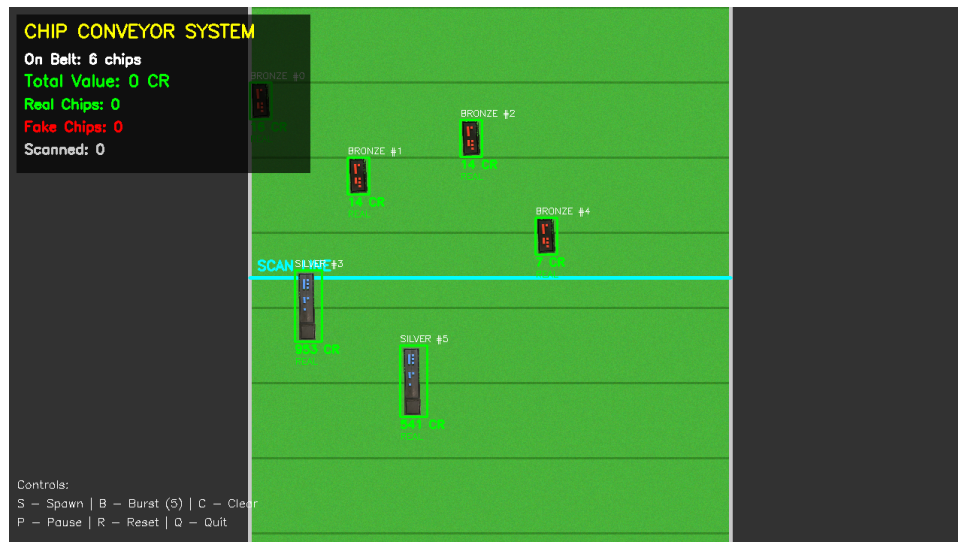
1. Empty Conveyor Belt



Empty Conveyor

The initial state showing the green conveyor belt (50% screen width, centered) with scan line and statistics panel.

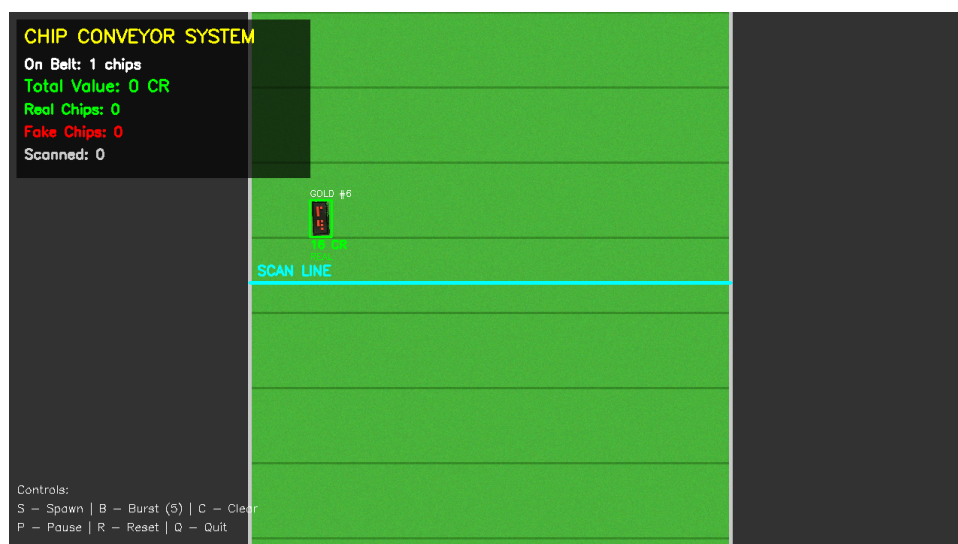
2. Multiple Chips in Motion



Multiple Chips

Several chips moving down the conveyor belt. Shows mix of Gold, Silver, and Bronze chips with real-time value calculation and statistics tracking.

3. Gold Chip Detection

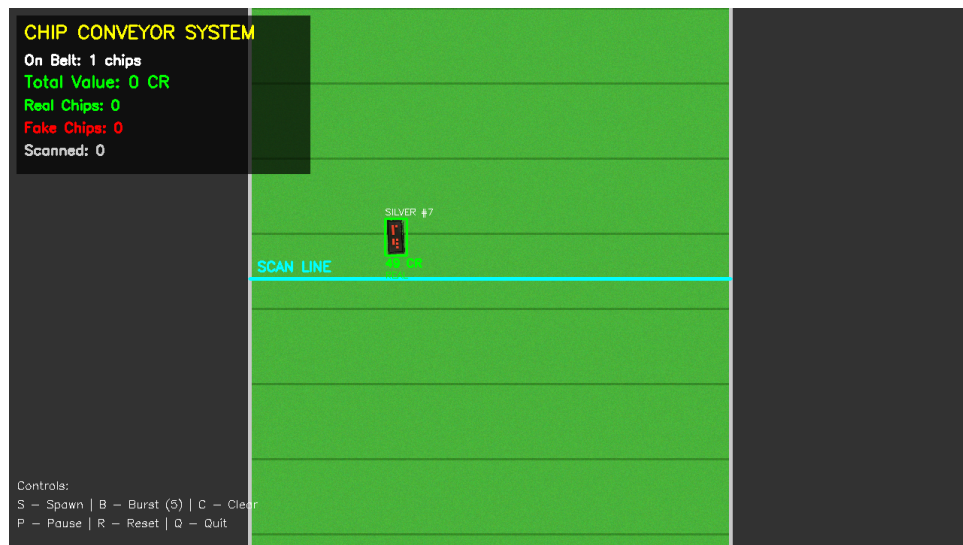


Gold Chip

Individual Gold chip with:

- Yellow color coding
- 3-digit number display
- Value calculation: digits $\times$ 10
- Real-time annotation

4. Silver Chip Detection

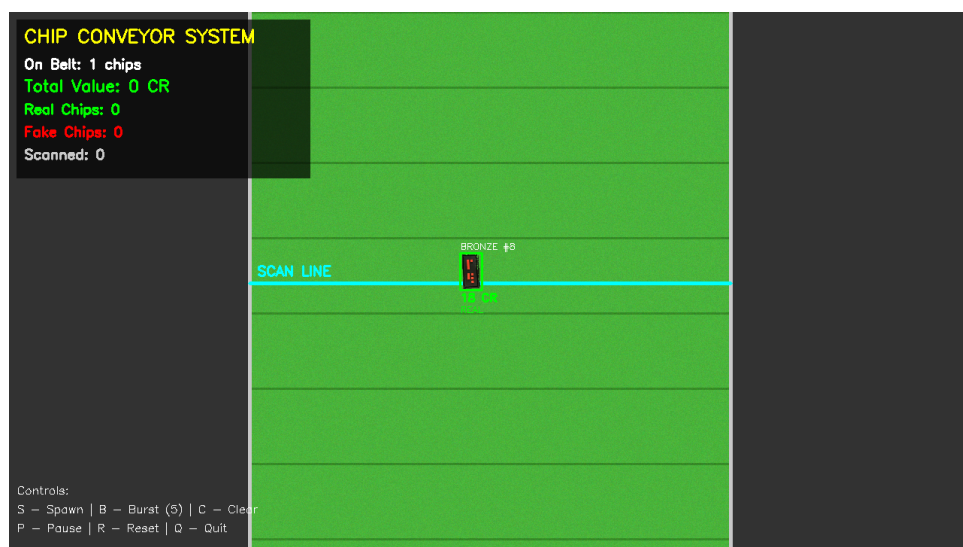


Silver Chip

Silver chip showing:

- Gray color coding
- Direct 3-digit value
- Bounding box in green (authentic)
- Value display in CR (Credits)

5. Bronze Chip Detection

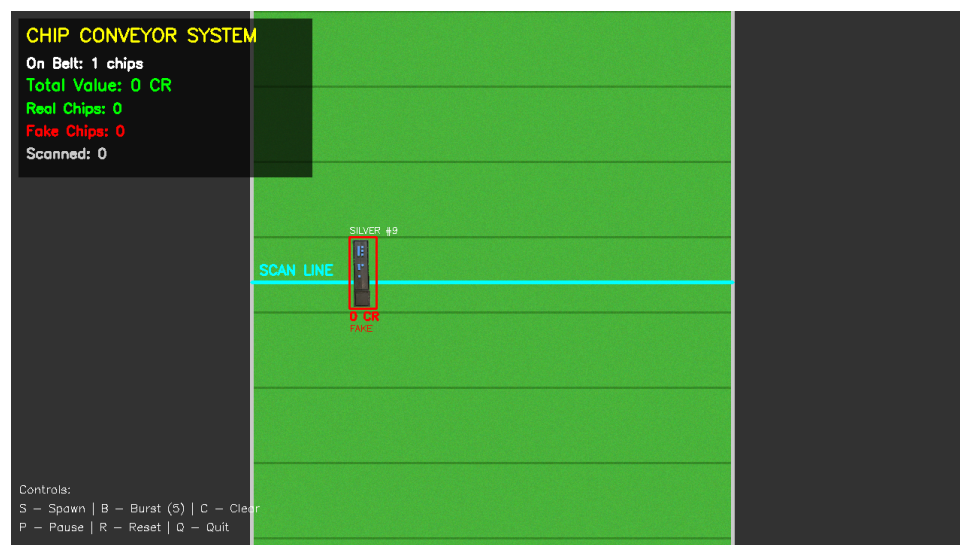


Bronze Chip

Bronze chip with:

- Orange color coding
- Multiplied digit value
- Position on conveyor belt
- Authenticity marker

6. Fake Chip Detection

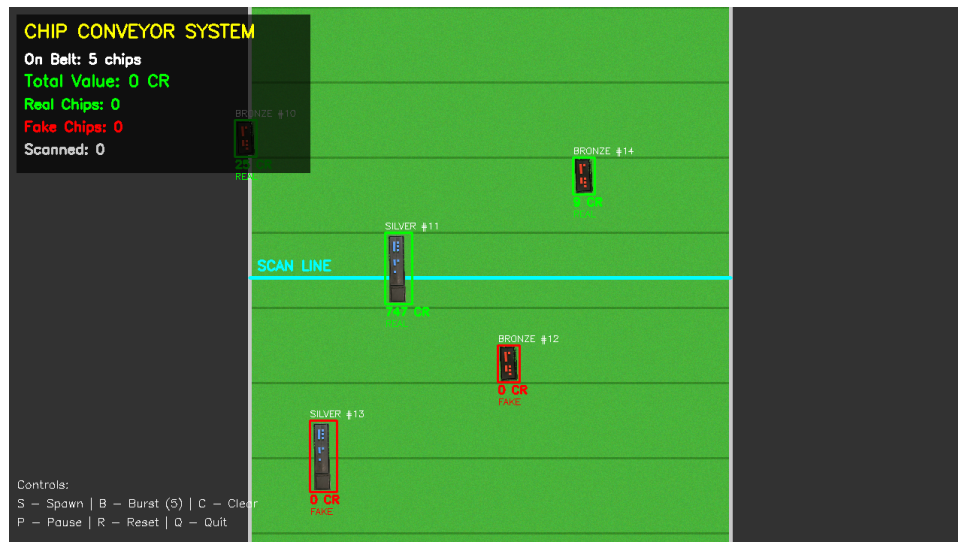


Fake Chip

Counterfeit chip identified by:

- Red bounding box
- "FAKE" label
- Zero value (0 CR)
- Not counted in total value

7. Full Simulation



Full Simulation

Complete system demonstration showing:

- Multiple chip types simultaneously
- Real-time statistics panel (top-left)
- Total value accumulation
- Real vs Fake count tracking
- Control instructions (bottom-left)
- Scan line in center
- FPS display

System Features Demonstrated

- **Color-coded Detection:** Gold (Yellow), Silver (Gray), Bronze (Orange), Fake (Red)
- **Real-time Processing:** 60 FPS in simulator mode
- **Value Calculation:** Automatic computation based on chip type
- **Statistics Tracking:** Live total value and chip counts
- **Visual Feedback:** Clear annotations and status indicators
- **Conveyor Physics:** Realistic perpendicular movement

Technical Details

- **Resolution:** 1280 × 720 pixels
- **Belt Width:** 640 pixels (50% of screen)
- **Processing:** OpenCV + NumPy
- **Color Space:** HSV for robust detection
- **Alpha Blending:** PNG transparency support

Project: STB600 Final Project 2025

Group: 8

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